

AL WAJH, Saudi Arabia

WMO: 404000

Lat: 26.199N

Lon: 36.476E

Elev: 66

StdP: 14.66

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	55.2	57.0	30.1	24.3	66.4	34.0	28.8	65.9	27.3	68.0	24.2	69.2	6.0	20	0.442

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.6	97.0	77.4	95.0	80.0	93.4	80.9	86.4	92.1	85.4	91.5	84.5	90.8	12.1	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
84.6	182.3	90.7	84.1	179.2	90.5	82.7	170.8	89.6	51.4	93.1	50.2	90.8	49.0	91.1	92.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.0	22.8	20.8	50.7	108.1	3.1	4.2	48.5	111.2	46.7	113.6	44.9	116.0	42.7	119.1		
(5)			41.2	88.3	3.2	1.7	39.0	89.5	37.1	90.5	35.3	91.4	33.0	92.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	78.1	67.2	68.7	71.9	76.5	81.5	83.7	86.3	86.9	84.6	81.7	76.4	70.5
	DBStd	7.73	3.95	4.11	4.60	4.52	4.28	3.66	2.35	2.33	2.71	3.23	3.93	3.93
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	33	19	9	2	0	0	0	0	0	0	0	0	3
	CDD50	10239	535	524	679	794	976	1012	1126	1144	1039	983	792	635
	CDD65	4796	88	113	216	344	511	562	661	679	589	518	342	173
	CDH74	56597	439	613	1458	3109	6063	7454	9556	9936	7867	6052	2991	1059
	CDH80	25716	41	84	304	914	2537	3573	5201	5544	3844	2552	937	185
(14) Wind	WSAvg	9.3	8.6	9.5	10.1	9.9	9.9	10.2	9.2	9.7	9.9	8.6	8.1	8.1
(15) Precipitation	PrecAvg	1.3	0.6	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.2	0.2
	PrecMax	3.1	2.3	2.0	0.4	0.2	0.2	0.0	0.0	0.0	0.1	0.7	1.5	2.2
	PrecMin	0.4	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	0.9	0.6	0.4	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.4	0.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	84.2	86.3	91.3	95.0	98.9	101.6	98.8	98.4	97.7	96.5	94.7	89.4
		MCWB	66.3	66.1	69.4	71.1	72.7	74.0	78.1	80.9	77.9	73.3	67.8	65.6
	2%	DB	80.5	82.3	86.1	90.0	94.6	95.2	95.3	93.6	91.8	89.5	84.2	
		MCWB	67.0	68.0	69.3	71.9	74.7	77.8	81.0	83.1	81.5	79.3	72.3	67.6
	5%	DB	78.6	79.8	82.7	87.5	91.3	92.9	93.6	93.9	91.8	90.0	87.1	81.8
		MCWB	66.1	67.5	69.8	71.7	74.7	79.1	81.7	83.2	81.9	79.4	72.5	67.6
	10%	DB	76.6	77.4	80.7	84.5	89.2	91.0	92.8	93.1	91.1	88.9	84.9	79.2
		MCWB	64.7	65.8	69.3	71.8	75.4	78.9	81.8	83.2	81.8	79.0	71.8	65.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.9	74.6	76.4	79.1	82.7	85.2	87.9	88.3	87.1	85.2	79.2	75.3
		MCDB	78.8	79.7	82.7	86.4	89.8	92.1	92.9	92.9	91.8	90.5	86.5	81.5
	2%	WB	70.8	72.2	74.3	77.3	80.9	83.6	86.2	86.7	85.4	83.3	77.3	72.4
		MCDB	77.4	78.2	81.2	84.5	88.1	90.5	92.0	92.4	90.9	88.9	85.0	79.9
	5%	WB	68.0	70.0	72.5	75.9	79.6	82.3	84.9	85.7	84.3	81.9	75.4	69.9
		MCDB	75.9	77.4	80.0	83.1	86.9	88.9	91.2	91.9	90.1	87.9	83.8	78.6
	10%	WB	65.5	67.6	70.7	74.5	78.5	81.1	83.9	84.7	83.2	80.5	73.7	67.7
		MCDB	74.8	76.2	78.8	81.9	85.9	88.1	90.4	91.1	89.1	87.2	82.6	77.5
(35) Mean Daily Temperature Range	5% DB	MDBR	17.0	16.9	16.8	15.6	14.5	14.1	13.5	12.6	13.6	15.0	17.3	17.5
		MCDBR	19.1	18.9	19.0	18.5	17.3	16.6	14.7	13.2	14.5	15.5	19.2	20.0
		MCWBR	14.4	15.8	16.2	16.2	15.6	14.0	11.5	10.5	12.4	15.2	15.5	14.7
	5% WB	MCDBR	16.7	16.4	16.6	15.3	14.5	14.1	13.5	12.3	13.4	14.5	17.2	17.3
		MCWBR	14.3	15.5	15.5	14.6	14.3	12.6	11.5	10.8	12.2	15.2	15.4	14.1
(40) Clear-Sky Solar Irradiance	taub		0.370	0.415	0.458	0.520	0.513	0.461	0.485	0.513	0.472	0.466	0.406	0.382
	taud		2.342	2.166	2.032	1.889	1.924	2.090	2.064	1.996	2.075	2.078	2.242	2.294
	Ebn at Noon		277	272	266	252	253	265	258	251	258	252	262	267
	Edh at Noon		35	45	54	64	62	52	53	57	51	48	38	35
(44) All-Sky Solar Radiation	RadAvg		1360	1676	2078	2341	2510	2633	2544	2385	2154	1787	1466	1279
	RadStd		76	60	59	67	81	29	42	43	52	42	44	30

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+1.10	N/A	N/A	+1.22	+1.08	+0.94	N/A	N/A	+416	+409
(47) Regional (0 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page